

SHD-CCP Protocol Calibration Sheet

Full Name: Symbolic High Dimensional Context Compression Protocol

Description: A 64-bit neural symbolic AI to AI communication standard language.

Protocol Version: 1.0

Encoding: Big-Endian Hexadecimal / Binary Stream

Core Latency Target: < 10ms (Local Assembly)

Integrity Check: CRC32 (Hardware Layer) + Falcon-Sign (Cryptographic Layer)

1. Logical Packet Frame (LPF)

CommonCore

The fundamental data unit encapsulated within all SCP packets.

Segment	Bit Width	Byte Size	Data Type	Description
Header	32 bits	4 Bytes	Metadata	[Ver(4b)][Type(4b)][Timestamp(24b)]
Payload	64 bits	8 Bytes	Core Token	[Dyn(8)][Scale(8)][Quaternion(32)][Scale(8)][Sym(8)]
Footer	32 bits	4 Bytes	Checksum	CRC32 of Header + Payload
TOTAL	128 bits	16 Bytes	Binary	Base frame for encryption

2. Secure Chain Packet (SCP) - Type: GENESIS

Root authority packet. Establishes the trust anchor for a new chain.

Transmission Specs (Falcon-512 Mode)

- Payload (LPF):** 16 Bytes
- Linkage (Public Key):** 897 Bytes (Polynomial coefficients)
- Authentication (Sig):** 666 Bytes (Falcon Signature of LPF+Key)

- **Total Packet Size: 1,579 Bytes**

Transmission Specs (Falcon-1024 Mode)

- **Payload (LPF):** 16 Bytes
- **Linkage (Public Key):** 1,793 Bytes
- **Authentication (Sig):** 1,280 Bytes
- **Total Packet Size: 3,089 Bytes**

Calibration Note: Genesis packets incur higher bandwidth overhead due to the embedding of the full Public Key. Transmission buffers must handle the larger `Linkage` block.

3. Secure Chain Packet (SCP) - Type: CHAINED

Standard data packet. Links cryptographically to the preceding packet.

Transmission Specs (Falcon-512 Mode)

- **Payload (LPF):** 16 Bytes
- **Linkage (Prev Sig):** 666 Bytes (Signature of $Packet_{n-1}$)
- **Authentication (Sig):** 666 Bytes (Falcon Signature of LPF+Linkage)
- **Total Packet Size: 1,348 Bytes**

Transmission Specs (Falcon-1024 Mode)

- **Payload (LPF):** 16 Bytes
- **Linkage (Prev Sig):** 1,280 Bytes
- **Authentication (Sig):** 1,280 Bytes
- **Total Packet Size: 2,576 Bytes**

Calibration Note: Chained packets represent the steady-state flow. Throughput calculations should primarily use the Chained Packet size. The `Linkage` field here acts as a "Hash Pointer" to the history, ensuring immutability.

4. Transmission Latency Benchmarks

Estimated transmission time per packet based on standard network speeds.

Network Speed	Genesis (512)	Chained (512)	Genesis (1024)	Chained (1024)
1 Gbps	12.6 μ s	10.7 μ s	24.7 μ s	20.6 μ s

100 Mbps	126 μs	107 μs	247 μs	206 μs
10 Mbps	1.26 ms	1.07 ms	2.47 ms	2.06 ms